Lecture plan · for the lecturer

Heuristics and simulated annealing

Optimization · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Use local search.
- Explain annealing.
- Evaluate heuristics.

Activity: Every row a seed

1. Warm-up (5 min): Ask students whether they would wait all weekend for a proven optimum or take a good timetable now.
2. Search (15 min): Run local search and see where it gets stuck.
3. Anneal (20 min): Each row runs annealing with its own seed and reports the final cost.
4. Evaluate (15 min): Plot all results and compare with the known optimum.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How wide was the spread?
- What did cooling faster do?
- How will you require evaluation?

Exit ticket

- Write the evaluation rule you will require for heuristics.

Try it in your next lecture
Give each row of students a different random seed and the same timetable problem; each row runs annealing and reports its final cost, then plot all results and discuss the spread.

Seminar sheet · for students

Heuristics and simulated annealing

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Name _____ Group _____ Date _____

1 **At T = 2, a move worse by 2 is accepted with probability about:**

◯ 100% ◯ 37% ◯ 14%

2 **Compute acceptance.**

A move worse by Δ = 3 at T = 3 is accepted with probability e^(_____), about _____ %.

Word bank: 5 · -1 · 37 · -3

3 **Which make a heuristic comparison fair? Choose all that apply.**

☐ Tuning on the comparison instances ☐ Reporting gaps to a lower bound

☐ Several seeds per method ☐ Equal time budgets

4 **Simulated annealing with a geometric cooling schedule is guaranteed to find the global optimum.**

◯ True ◯ False

5 **Match each temperature to the search behaviour.**

Very high T	• • • •	Explores but prefers better moves
Moderate T	• • • •	Behaves like pure local search
T near zero	• • • •	Close to a random walk

Answers · for the lecturer

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1 **37%**

e^(−2/2) = e^(−1) ≈ 0.37.

2 **A move worse by Δ = 3 at T = 3 is accepted with probability e^(−1), about 37 %.**

e^(−3/3) = e^(−1) ≈ 0.37.

3 **Several seeds per method; Equal time budgets; Reporting gaps to a lower bound**

Tuning on the test instances overfits, just as in machine learning.

4 **False**

The convergence result needs very slow logarithmic cooling; practical schedules carry no guarantee.

5 **Very high T — Close to a random walk**
Moderate T — Explores but prefers better moves
T near zero — Behaves like pure local search

Cooling moves the search from exploration to refinement.

For you to think about

How do your students currently report heuristic results, and what would convince you a result is good?